

# EchoVerse

Real-Time Bidirectional AI Voice Agent

STT · LLM · TTS · RAG · WebRTC · WebSockets

## OVERVIEW

- Users upload notes or paste any content
- System narrates the content in audio form
- User can interrupt anytime using voice
- System answers based on context (RAG)
- Narration resumes intelligently after Q&A
- Real-time conversational experience
- Interruptible narration
- Context-aware answering via RAG
- Production-ready modular architecture
- Fully deployable with public URL

■ Goal: Simulate "group study with AI" — a living, breathing study partner that reads for you and answers your doubts.

## TECHNOLOGY

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| <div>◆ Frontend</div> <div>■ React</div> <div>■ TailwindCSS</div> <div>■ WebRTC</div> | <div>◆ Backend</div> <div>■ FastAPI</div> <div>■ WebSockets</div>                     | <div>◆ STT</div> <div>■ Whisper (local — zero cost)</div>       |
| <div>◆ LLM Brain</div> <div>■ Groq (low-latency primary)</div>                        | <div>◆ TTS</div> <div>■ ElevenLabs (primary)</div> <div>■ Google TTS (fallback)</div> | <div>◆ Database</div> <div>■ PostgreSQL (Neon / Supabase)</div> |
| <div>◆ Vector DB</div> <div>■ FAISS (local, fast RAG)</div>                           |                                                                                       |                                                                 |

## DESIGN

|    |                             |   |
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| 01 | User (Mic Input)            | ▼ |
| 02 | WebRTC Frontend             | ▼ |
| 03 | FastAPI Backend (WebSocket) | ▼ |

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| 04 | Whisper STT                      | ▼ |
| 05 | RAG — FAISS Retrieval            | ▼ |
| 06 | Groq LLM                         | ▼ |
| 07 | TTS — ElevenLabs / Google        | ▼ |
| 08 | Audio Stream → Frontend Playback | ✓ |

■ WORKFLOW

|                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                     |
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| <p>■ <b>Narration Flow</b></p> <ol style="list-style-type: none"><li>1■■ User uploads text</li><li>2■■ Backend splits into chunks</li><li>3■■ LLM optionally refines text</li><li>4■■ TTS generates audio</li><li>5■■ Audio streamed to frontend</li></ol> | <p>■■ <b>Interrupt Flow</b></p> <ol style="list-style-type: none"><li>1■■ User speaks mid-narration</li><li>2■■ Audio sent in real-time chunks</li><li>3■■ Silence detection triggers STT</li><li>4■■ Whisper: speech → text</li><li>5■■ RAG retrieves context</li><li>6■■ Groq LLM generates answer</li><li>7■■ TTS speaks the answer</li><li>8■■ Narration auto-resumes</li></ol> |
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■ FEATURES

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| <p>■■ <b>Core</b></p> <ul style="list-style-type: none"><li>✓ Real-time narration</li><li>✓ Voice interruption</li><li>✓ AI Q&amp;A (RAG-based)</li><li>✓ Auto-resume narration</li></ul> | <p>■ <b>Intelligence</b></p> <ul style="list-style-type: none"><li>✓ Context-aware answers</li><li>✓ Conversation memory</li><li>✓ Code explanation mode</li><li>✓ Interactive study mode</li></ul> | <p>■ <b>Audio</b></p> <ul style="list-style-type: none"><li>✓ High-quality voice (ElevenLabs)</li><li>✓ Google TTS fallback</li><li>✓ Low-latency &lt;3s response</li></ul> | <p>■■ <b>System</b></p> <ul style="list-style-type: none"><li>✓ Multi-provider architecture</li><li>✓ Graceful failover</li><li>✓ Usage limits &amp; quotas</li><li>✓ Modular pipeline</li></ul> |
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■ ENGINEERING

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| <pre>backend/<br/>■■■ main.py<br/>■■■ routes/<br/>■ ■■■ audio.py<br/>■ ■■■ llm.py<br/>■ ■■■ tts.py<br/>■■■ services/<br/>■ ■■■ whisper_service.py<br/>■ ■■■ rag_service.py<br/>■ ■■■ groq_service.py<br/>■ ■■■ tts_service.py<br/>■■■ utils/<br/>■ ■■■ audio_utils.py<br/>■ ■■■ vad.py<br/>■■■ models/<br/>■ ■■■ user.py<br/>■ ■■■ session.py<br/>■■■ config/<br/>■■■ settings.py</pre> | <div>■ Required API Keys</div> <pre># .env file<br/><br/>GROQ_API_KEY=gsk_...<br/><br/>ELEVENLABS_API_KEY=...<br/><br/>ELEVENLABS_VOICE_ID=...<br/><br/>GOOGLE_TTS_API_KEY=...<br/><br/>DATABASE_URL=postgresql://...</pre> |
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■ RISKS & CONSTRAINTS

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| <div>■■ Key Decisions</div> <ul style="list-style-type: none"><li>■ Whisper locally — Zero STT cost, works on CPU</li><li>■ Groq as LLM — Lowest latency among providers</li><li>■ ElevenLabs TTS — Best voice quality available</li><li>■ Google TTS fallback — Reliability &amp; cost control</li><li>■ FAISS local — Fast RAG without cloud dependency</li></ul> | <div>■ Constraints</div> <ul style="list-style-type: none"><li>■■ CPU only — no GPU dependency</li><li>■■ Audio chunks: 30–60 sec limit</li><li>■■ Graceful API failure handling</li><li>■■ &lt;3 sec end-to-end latency target</li></ul> | <div>■ Risk Areas</div> <ul style="list-style-type: none"><li>■ Whisper latency on slow CPUs</li><li>■ ElevenLabs credit exhaustion</li><li>■ WebRTC streaming edge cases</li><li>■ Audio stream sync issues</li></ul> |
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■ DEPLOYMENT

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| <div>■ <b>Deployment Targets</b></div> <div>Frontend → <b>Vercel</b></div> <div>Backend → <b>Render / Fly.io</b></div> <div>■ <b>Testing Checklist</b></div> <div><div>■ Narration flow end-to-end</div><div>■ Voice interruption accuracy</div><div>■ TTS fallback trigger</div><div>■ Deployment stability</div></div> | <div>■ <b>Future Enhancements</b></div> <div><div>■ Multi-speaker voices</div><div>■ Emotion detection in voice</div><div>■ Real-time streaming TTS</div><div>■ Mobile app (iOS / Android)</div><div>■ Subscription model</div></div> | <div>■ <b>Final Deliverable</b></div> <div><div>■ Live deployed URL</div><div>■ Working voice interaction</div><div>■ Stable narration + interruption</div><div>■ Clean, polished UI</div></div> <div>■ <b>What This Is</b></div> <div><div>■ NOT an audiobook generator</div><div>■ <b>A real-time conversational AI voice agent with full bidirectional interaction</b></div></div> |
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